

Automatic Detection of Residential Buildings Using LIDAR Data and Multispectral Imagery

Mohammad Awrangjeb*, Mehdi Ravanbakhsh and Clive S. Fraser

Cooperative Research Centre for Spatial Information, The University of Melbourne

723 Swanston St, Carlton Vic 3053, Australia

Phone: +61 3 8344 9182, Fax: +61 3 9349 5185

Email: {mawr, m.ravanbakhsh, c.fraser}@unimelb.edu.au

Abstract

This paper presents an automatic building detection technique using LIDAR data and multispectral imagery. Two masks are obtained from the LIDAR data: a ‘primary building mask’ and a ‘secondary building mask’. The primary building mask indicates the *void areas* where the laser does not reach below a certain height threshold. The secondary building mask indicates the *filled areas*, from where the laser reflects, above the same threshold. Line segments are extracted from around the void areas in the primary building mask. Line segments around trees are removed using the normalized difference vegetation index derived from the orthorectified multispectral images. The initial building positions are obtained based on the remaining line segments. The complete buildings are detected from their initial positions using the two masks and multispectral images in the YIQ colour system. It is experimentally shown that the proposed technique can successfully detect urban residential buildings, when assessed in terms of 15 indices including

*Corresponding author

completeness, correctness and quality.

Key words: Building detection, LIDAR, point cloud, multispectral, photogrammetry imagery, orthoimage, fusion, edge feature

1. Introduction

Building detection from remotely sensed data is important to the real estate industry, city planning, homeland security, disaster (flood or bush fire) management and many other applications. The automated extraction of building boundaries is also a crucial step towards generating city models (Cheng et al., 2008). Consequently, a large number of building detection techniques have been reported over the last few decades.

However, 100% successful automatic building detection is still an unrealized goal. There are several reasons to explain this situation (Sohn and Dowman, 2007). These include:

- *Scene complexity*: most of the scenes usually contain very rich information which provides a large number of cues with geometric or chromatic co-similarity to buildings, but belong to non-building objects.
- *Incomplete cue extraction*: there is always a significant loss of relevant building cues due to occlusions, poor contrast, shadows and disadvantageous image perspective.
- *Sensor dependency*: the primary data to support the building detection is available from a variety of sources with different resolution, each source having its own advantages and disadvantages for building detection.

21 Lee et al. (2008) have categorized building detection techniques into three
 22 groups. Firstly, there are many algorithms which use 2D or 3D information
 23 from photogrammetric imagery (Mayer, 1999). The complexity of separating
 24 buildings from other objects increases with the increase of image resolution
 25 as high-resolution images contain more detailed information (Cheng et al.,
 26 2008), along with occlusions and shadows (Yong and Huayi, 2008). The
 27 derivation of 3D information, for example, the depth information from stereo
 28 by multiple images (Sun et al., 2005), is even more complicated (Vu et al.,
 29 2009). In addition, nearby trees of similar height also make the use of such
 30 derived range data difficult (Lee et al., 2008).

31 Secondly, there have been several attempts to detect building regions from
 32 LIDAR (LIght Detection And Ranging) data. This task has been largely
 33 solved by classifying the LIDAR points according to whether they belong
 34 to bare-earth, buildings, or other object classes (Lee et al., 2008). In fact,
 35 the introduction of LIDAR has offered a favourable option for improving
 36 the level of automation in the building detection process when compared
 37 to image-based detection (Vu et al., 2009). Oude Elberink (2008) has dis-
 38 cussed a number of problems with building detection using LIDAR data and
 39 it has been shown that the use of raw or interpolated data can influence
 40 the detection performance (Demir et al., 2009). Moreover, there may be
 41 poor horizontal accuracy for building edges (Yong and Huayi, 2008) and it
 42 is hard to obtain a detailed and geometrically precise boundary using only
 43 LIDAR point clouds (Cheng et al., 2008). The quality of regularized building
 44 boundaries also depends on LIDAR resolution (Sampath and Shan, 2007).

45 LIDAR and photogrammetric imagery each have particular advantages

46 and disadvantages in horizontal and vertical positioning accuracy. Compared
47 with photogrammetric imagery, LIDAR generally provides more accurate
48 height information but less accurate boundary lines. Unfortunately, some
49 regions in LIDAR data have null values due to self-occlusion of a building
50 or if they contain water. Photogrammetric imagery can provide extensive
51 2D information such as high-resolution texture and colour information as
52 well as 3D information from stereo images. As a result, several authors have
53 promoted an integration of LIDAR data and imagery as a means of advancing
54 building detection (Rottensteiner et al., 2005; Yong and Huayi, 2008; Cheng
55 et al., 2008; Demir et al., 2009).

56 The third category of methods does use both LIDAR data and pho-
57 togrammetric imagery. More specifically, intensity and height information
58 in LIDAR data can be used with texture and region boundary information
59 in aerial imagery to improve accuracy (Lee et al., 2008).

60 However, the question of how to integrate the two data sources for build-
61 ing boundary extraction still arises; few approaches with technical details
62 have thus far been published (Rottensteiner et al., 2005). The question of
63 how to combine the two different data sources in an optimal way so that
64 their weaknesses can be compensated effectively is an active area of current
65 research (Yong and Huayi, 2008).

66 Regarding performance evaluation, there is a current lack of uniform and
67 rigorous evaluation systems, and an absence of standards (Rutzinger et al.,
68 2009). Indeed, evaluation results are often missing from published accounts
69 of building detection (Yong and Huayi, 2008); the use of 1 to 2 evaluation
70 indices only has characterized many studies (Demir et al., 2009; Vu et al.,

71 2009).

72 This paper aims at following two goals: a successful integration of the
73 LIDAR data and photogrammetric imagery for building detection so that the
74 increased detection performance is obtained and development of an automatic
75 performance evaluation system using 15 evaluation indices.

76 The proposed automatic building detection technique uses raw LIDAR
77 data and orthoimagery. Two masks are obtained from the LIDAR data: a
78 ‘primary building mask’ and an ‘secondary building mask’. Line segments
79 around the black shapes (absence of height data) in the primary building
80 mask constitute the initial building positions. The final buildings are then de-
81 tected extending their initial positions using the multispectral images, trans-
82 formed into the YIQ (intensity, hue and saturation) colour system. The two
83 masks ensure accurate delineation of the buildings. In particular, the pri-
84 mary building mask helps separate detected buildings when they are very
85 close to each other and the secondary building mask helps to avoid exten-
86 sions to initial positions outside a building when the roof and ground have
87 similar colour information. It is experimentally shown that the proposed
88 technique can detect rectilinear buildings with a favourable success rate, es-
89 pecially within the Australian urban environment for which it was primarily
90 developed.

91 The proposed detection technique has similarities to that reported by
92 Sohn and Dowman (2007) and Cheng et al. (2008) in the sense that it uses
93 line segments and a regularization step (adjustment) employing dominant
94 line angles.

95 The proposed automatic evaluation system uses both object- and pixel-

based indices. In addition, though the pixel-based evaluation indirectly reflects the horizontal accuracy, the geometric evaluation is introduced as a means of direct estimation of the horizontal, geometric or positional accuracy.

The performance of the proposed building detection approach has been evaluated through using 15 indices in three categories, these being object-based, pixel-based and geometric. Most of the indices have been adopted from the literature and the remainder are proposed here for a more complete evaluation.

The rest of the paper is organized as follows: Section 2 presents a review of both existing integration techniques for photogrammetric imagery and LIDAR data for both building detection and performance evaluation systems. Section 3 details the proposed building detection technique. The proposed evaluation system with experimental test results is discussed in Section 4. Finally, concluding remarks are offered in Section 5.

2. Related Work

2.1. Integration of LIDAR and Imagery

Building detection techniques integrating LIDAR data and imagery can be divided into two groups. Firstly, there are techniques which use the LIDAR data as the primary cue for building detection and employ the imagery only to remove vegetation (Rottensteiner et al., 2005; Vu et al., 2009). As a result, they suffer from poor horizontal accuracy for the detected buildings. Rottensteiner et al. (2005) employed the Dempster-Shafer theory as a data fusion framework to classify points as buildings, trees, grassland or bare soil.

120 However, the detection performance was adversely affected for small build-
121 ings (Rottensteiner et al., 2007). The reason is that if the Dempster-Shafer
122 model is not properly trained, then the miss-classification rate increases con-
123 siderably (Khoshelham et al., 2008). Vu et al. (2009) used a morphological
124 scale-space for extracting building footprints from the elevation data and
125 then removed vegetation areas using the spectral data. The detection per-
126 formance was low and high computational complexity was reported because
127 of using the scale-space.

128 Secondly, there are integration techniques (Haala and Brenner, 1999;
129 Chen et al., 2004; Sohn and Dowman, 2007; Lee et al., 2008; Demir et al.,
130 2009) which use both the LIDAR data and the imagery as the primary cues
131 to delineate building outlines. They also employ the imagery to remove veg-
132 etation. Consequently, they offer better horizontal accuracy for the detected
133 buildings. The proposed building detection technique falls into this group.
134 Haala and Brenner (1999) applied a pixel-based classification where the nor-
135 malized DSM (nDSM) was used as an additional channel to the three spec-
136 tral bands of the aerial imagery. Chen et al. (2004) followed a region-based
137 segmentation of nDSM and orthoimages and then used a knowledge-based
138 classification to detect building. However, this method did not show how to
139 cope with erroneous lines (Sohn and Dowman, 2007) and could not detect
140 small buildings.

141 Sohn and Dowman (2007) employed a data-driven approach on the optical
142 imagery and a model-driven approach on the point cloud to extract rectilin-
143 ear lines around buildings. Extracted lines were regularized by analyzing the
144 dominant line angles. Cheng et al. (2008) proposed a similar technique with

precise geometric position. Lee et al. (2008) extracted the initial building boundaries from the LIDAR data and then enhanced the initial boundaries using colour information, after which edge matching and perceptual grouping techniques were applied to yield the final building boundaries. Demir et al. (2009) applied four different methods to achieve an improvement by combining the advantages and disadvantages of these approaches and used the edge information from images for quality improvement of the detected buildings.

2.2. Evaluation Systems

Performance evaluation systems reported in the literature can be divided into two groups: those using overlapping thresholds (Rottensteiner et al., 2005; Rutzinger et al., 2009; Lee et al., 2008) and those not using any thresholds (Shan and Lee, 2005; Shufelt, 1999). Threshold-based systems use one or more overlapping thresholds while making correspondences between detected and reference building sets. The problem with threshold-based systems is that they are subjective and likely to be controversial since there is no unique way to select the thresholds (Shufelt, 1999).

The evaluation systems can also be categorized into pixel-based systems (Rottensteiner et al., 2005; Rutzinger et al., 2009; Lee et al., 2008) and object-based systems (Rutzinger et al., 2009). While the latter counts the number of buildings and offers a quick assessment, the former is based on the number of pixels and provides more rigorous evaluation (Song and Haithcoat, 2005). The pixel-based evaluation indirectly corresponds to the horizontal accuracy of the detected building footprints.

In Rottensteiner et al. (2005) and Rutzinger et al. (2009), a correspon-

170 dence was established between a detected building and a reference building
171 if they overlapped each other either strongly, more than 80% overlap, or par-
172 tially, 50% to 80% overlap. Both of the above evaluation systems do not
173 reflect the actual detection scenario. Firstly, the presence of *false positive*
174 and *false negative* detections is not considered at all. Secondly, there may
175 be many-to-many relationships between the detected and reference sets and
176 such relationships are considered as error (Shan and Lee, 2005). Finally,
177 merging and splitting of the detected buildings as in Rutzinger et al. (2009)
178 does not necessarily correspond to the actual performance.

179 Without using a particular overlapping threshold, Shufelt (1999) showed
180 the detection performance graphically as the overlapped area varied from
181 0-100%. Shan and Lee (2005) presented results by histograms showing the
182 frequency of buildings as functions of *underlap*, *overlap*, *extralap*, *crosslap*,
183 and *fitness*. The number of *false negative* buildings was indicated by the
184 frequency at 100% *underlap* and the number of *false positive* buildings was
185 indicated by the frequency both at *crosslap* 0 and 0% *fitness*.

186 3. Proposed Building Detection Technique

187 The proposed automatic building detection technique uses LIDAR data
188 and colour orthoimagery. It has four major steps. Firstly, two masks, a
189 ‘primary building mask’ and an ‘secondary building mask’, are generated
190 from the LIDAR data. The primary building mask indicates the *void areas*
191 where there are no laser returns below a certain height threshold. The sec-
192 ondary building mask indicates the *filled areas*, from where returns indicate
193 an elevated object above the same height threshold. Secondly, line segments

194 from around the void areas in the primary building mask are extracted. Line
 195 segments around trees are removed using Normalized Difference Vegetation
 196 Index (NDVI) values derived from the multispectral images. Thirdly, ini-
 197 tial building positions are recovered based on the remaining line segments.
 198 Finally, the complete building footprints are obtained from their initial po-
 199 sitions using the two masks and the orthoimagery in the YIQ colour system.

200 *3.1. Overview*

201 Fig. 1 shows the flow diagram of the proposed building detection tech-
 202 nique. The input information consists of a LIDAR point cloud, a DEM
 203 (digital elevation model) and multispectral orthoimagery. The point cloud
 204 and orthoimagery are registered to each other before being used as inputs.
 205 The primary and secondary building masks are first derived from the LIDAR
 206 data, along with NDVI values from the orthoimagery. The initial building
 207 positions are derived from the primary building mask. The colour informa-
 208 tion in the multispectral images is usually in the RGB system and therefore is
 209 converted into the YIQ system. The final buildings are obtained by extending
 210 their initial positions using the two masks and the YIQ colour information.

211 The following subsections detail the proposed detection technique. Sec-
 212 tion 4.4.1 presents the sensitivity analysis of important parameters used by
 213 the detection algorithm.

214 *3.2. Height Threshold and Masks*

215 While the primary building mask M_p is used for obtaining the initial
 216 building positions as rectangular areas, the secondary building mask M_s is

217 used as an indication of a maximum building size around an initial building
 218 position during the detection of a final building from its initial position.

219 All pixels in M_p are initially assigned 0 (false), but in M_s are assigned
 220 1 (true). The two masks are derived simultaneously by first dividing M_p
 221 (and hence M_s) into tiles of size 450×450 image pixels, since there may be
 222 different representative ground heights H_g in a large area. H_g is calculated
 223 separately for each tile from the corresponding DEM data. The DEM and
 224 LIDAR data are also grouped following the tiles of M_p . For each tile, H_g is
 225 simply estimated as the average of the height data from the corresponding
 226 DEM. Fig. 2(a)-(b) shows the tiles of masks on an orthoimage and the groups
 227 of LIDAR data.

228 In order to obtain the masks for each tile, a threshold $T_h = H_g + 2.5m$
 229 is applied. If the LIDAR height of a point (x, y) is less than T_h , the cor-
 230 responding pixel in M_p is assigned 1. If the height is greater than T_h , the
 231 corresponding pixel in M_s is assigned 0. In addition, since the horizontal
 232 resolution of LIDAR data is generally lower than that of the orthoimage, all
 233 the pixels in a 5×5 neighbourhood of (x, y) are also assigned 1 for M_p or 0
 234 for M_s . The size of the neighbourhood can be adjusted based on the relative
 235 resolutions of the LIDAR data and the orthoimage.

236 From Fig. 2(c) it can be seen that the majority of the buildings are distin-
 237 guishable in the primary building mask. However, when buildings are very
 238 close to each other, many are not clearly distinguishable in the secondary
 239 building mask, as shown in Fig. 2(d). Colour information from the orthoim-
 240 agery is therefore used for more accurate detection of the buildings (Section
 241 3.4).

242 3.3. Initial Building Positions

243 Initial building positions $B_{ini} = \{b_{ini,i}\}$, $1 \leq i \leq m$, where m is the num-
244 ber of detected positions, are detected as rectangular patches from the pri-
245 mary building mask M_p , with the black areas in Fig. 2(c) being the initial
246 building positions. This section describes how each of those areas is detected
247 as a rectangle or as a combination of two or more rectangles.

248 Three steps are followed to obtain B_{ini} from M_p . Firstly, lines around
249 the black shapes from M_p are formed. Secondly, the lines are adjusted and
250 extended. Finally, rectangular shapes are obtained using these lines.

251 3.3.1. Line Detection

252 The Canny edge detector (Canny, 1986) is first used to find all the edges
253 in M_p and then the short edges are discarded. Edges of less than $3m$ (20
254 pixels) in length are considered short, assuming that the minimum building
255 length or width is $3m$.

256 Since there may be noise and local variation introduced by the neighbour-
257 hood filling technique during the mask generation phase, a Gaussian kernel
258 with scale $\sigma = 3$ is utilized to smooth each edge.

259 Corners (absolute curvature maxima points) are then detected on each
260 of the smoothed curves using a fast corner detector described in Awrangjeb
261 et al. (2009). The smoothed curves are then decomposed into line segments.
262 On each edge, all the pixels between two corners or a corner and an endpoint
263 or two endpoints when enough corners are not available, are considered as
264 separate line segments. Again, short line segments, whose lengths are less
265 than $3m$, are discarded.

266 The detected corners and edge endpoints may not be well localized to
267 the building corners. In order to align the detected line segments with the
268 building edges, a least-squares straight-line fitting technique is applied. With
269 each line segment a point P_{in} is recorded. This ‘inside-point’ indicates on
270 which side of the line the building is recorded.

271 Obviously, some line segments around trees are obtained and in order to
272 avoid further processing of these, a rectangle of $3m$ width on the building
273 side is formed. The sigma of the NDVI value Υ inside the rectangle is then
274 employed, such that for a line segment if the mean of Υ is above a threshold
275 $T_{ndvi} = 48$, the line segment is classed as a tree-edge and removed. After the
276 application of the NDVI threshold on the extracted lines, the removed lines
277 are shown in Fig. 3(a) with circles at their centres.

278 It was found that the NDVI did not have a high discriminating power.
279 Many of the tree-edges could not be removed using T_{ndvi} . If a low T_{ndvi} value
280 was applied these tree-edges could be removed, but many important line seg-
281 ments which indicate the initial positions of buildings were also removed. A
282 similar effect was reported by Rottensteiner et al. (2007) who applied a post-
283 classification technique to improve the performance. Consequently, in this
284 investigation a minimum building length threshold ($3m$) has been applied to
285 the extracted line segments to remove small vegetation areas. In addition,
286 assuming that the buildings and their sides are locally parallel or perpendic-
287 ular to each other, the line segments are adjusted as discussed below. This
288 adjustment procedure removes a tree-edge which is neither locally parallel
289 nor perpendicular when compared to its neighbouring line segments.

290 3.3.2. *Adjusting and Extending Lines*

291 Under the assumption that the longer lines are more likely to be build-
292 ing edges, the extracted lines are sorted based on their lengths. Then, in
293 an iterative procedure starting from the longest line l_i and taking it as a
294 reference, the angle between the reference l_i and each line l_j in its neigh-
295 bourhood is estimated. A circular neighbourhood around the centre of l_i is
296 then considered. The radius of this neighbourhood is set as the maximum
297 building length, $50m$ in this investigation. If l_i and l_j are either parallel or
298 perpendicular to each other, to within a $\frac{\pi}{8}$ angular difference, the rotation
299 angle θ_r for l_j is estimated.

300 There may be buildings of different orientations in an area. This means
301 that one building or a group of buildings may have a different orientation
302 when compared to others in the neighbourhood. In order to avoid wrong
303 adjustments of the extracted lines, the lowest rotation angle θ_r is recorded
304 for each l_j over all iterations.

305 The above iterative procedure may be optionally terminated after a sig-
306 nificant number of iterations, say 50% of the number of total extracted lines.
307 After the iterative procedure, each l_j and its P_{in} are rotated with respect to
308 the line centre by its recorded angle θ_r . If a rotation angle is not recorded
309 for l_j , then this l_j is removed as a tree-edge. After the above adjustment
310 procedure, the removed lines are shown in Fig. 3(b) with circles at their
311 centres.

312 Each of the adjusted line segments may not represent a complete side
313 of a building. The line may be disrupted by trees, noise introduced in the
314 edge detection process, and by the neighbourhood filling effect. Therefore,

315 both ends of each adjusted line C_1C_2 are extended by considering a rectangle
 316 of length $L_e = 3m$ and width $W_e = 3m$ on each side (see Fig. 4(a) for C_2).
 317 Inside the rectangle the percentage of black pixels in M_p , Ω , should be high
 318 and more than 70%, and the mean Υ should be low at less than 48. This
 319 extension process continues iteratively and if any of the conditions fail at any
 320 iteration, W_e is halved. The process stops when W_e is less than the successive
 321 LIDAR point distance (i.e., $\leq 0.4m$ in this case).

322 3.3.3. Initial Buildings

323 Since long line segments represent more accurate building edges than
 324 short ones, the extended line segments are sorted again in descending order
 325 based on their lengths. In an iterative procedure, an initial building position
 326 is detected using the first longest line segment, another using the second
 327 longest line segment and so on. The rectangular positions are recorded in
 328 a set $B_{ini} = b_{ini,i}$, where $0 \leq i \leq m$, of four-points, one for each corner of a
 329 rectangle. B_{ini} is initially empty. Before detecting a rectangle using a line
 330 segment C_1C_2 in each iteration, C_1C_2 is tested to ascertain whether it is
 331 already in a detected rectangle $b_{ini,i}$.

332 In order to detect a new rectangle using C_1C_2 , an initial rectangle C_1C_2NM ,
 333 with length $L_b = |C_1C_2|$ and width $W_b = 1.5m$, is formed on the building side.
 334 Then three sides MN , C_1M and C_2N of C_1C_2NM are extended outwards
 335 with respect to P_{in} (Fig. 4(b)) using the same technique as that applied to
 336 extend the extracted lines, as discussed above.

337 After extension of three sides, if any of the sides of C_1C_2NM is not at
 338 least $3m$, C_1C_2 is removed as a tree-edge. Fig. 3(c) shows the initial building
 339 positions.

340 3.4. Final Building Positions

341 The final building positions are obtained from their initial positions by
342 extending each of the four sides. Image colour information and the two masks
343 M_p and M_s are considered during the extension. The colour information is
344 basically used to extend the initial positions; M_p is used to avoid unexpected
345 extension of an initial position over more than one actual buildings, and M_s
346 is used to avoid unexpected extension of an initial position beyond the actual
347 building roof.

348 In practice, there are different shapes of rectilinear buildings. We have
349 adopted a definition whereby a simple rectangular building (or building-part)
350 is termed an ‘I’ shape. Two adjoining perpendicular ‘I’ shapes then form
351 either an ‘L’ or ‘T’ shape building, whereas three connected rectangular
352 building parts form a ‘U’ shape, and four connected rectangular parts around
353 an open central area are termed a ‘C’ shape building.

354 If there are different rectangular initial positions for the same building, it
355 could be for one of the following two reasons. Firstly, an ‘I’ shape building
356 may be detected more than once. Secondly, the building is ‘L’, ‘T’, ‘U’ or ‘C’
357 shaped. In both of the above cases, the initial positions may overlap partly
358 or fully before or after their extensions. While in the first case, an overlap is
359 unexpected and has a negative impact in the detection performance, in the
360 second case an overlap is considered as a *natural overlap* and is expected to
361 join different detected parts of the same building, if necessary in any later
362 applications.

363 However, it is hard to decide which overlap is unexpected and which
364 is natural. If an initial building is completely within an already extended

building or building part, it is removed assuming that it is an unexpected overlap. Otherwise, it is extended assuming that it is a natural overlap. Before extending initial positions to obtain final positions, a preprocessing step is executed.

3.4.1. Preprocessing Initial Buildings

An initial building position may go outside the actual building roof due to a misregistration between the orthoimage and the LIDAR data. In order to avoid this, since the initial position will be extended outwards while obtaining the final position, its *length* and *width* are reduced by 15% before extension. For each reduced building position $ABCD$, the dominant colour threshold pairs $T_Y = [l_Y, h_Y]$, $T_I = [l_I, h_I]$ and $T_Q = [l_Q, h_Q]$ are estimated for intensity Y , hue I and saturation Q , respectively. Each dominant colour threshold pair indicates a range denoted by its *low* l and *high* h values.

In order to find threshold pairs for each band ($Y \in [0, 1]$, $I \in [-0.5957, 0.5957]$ and $Q \in [-0.5226, 0.5226]$), a histogram is generated for its values within $ABCD$, over 10 bins for Y or 20 bins for I and Q . Fig. 5 shows different types of histograms. The uphill and downhill histograms in Figs. 5(a)-(b) are two basic histograms and, practically, a histogram of Y , I or Q is a combination of these two. Fig. 5(c) shows one of the simple combinations that occurred most frequently in the experiments conducted for this investigation.

The histogram is divided into n parts, where $n \geq 1$. The value of n is usually 1, but may be greater than 1 if more colours appear on the building roof. Each part is from a minimum frequency bin to the next minimum frequency bin, or from a minimum frequency bin to the next maximum frequency bin, or from a maximum frequency bin to the next minimum frequency bin, if

390 enough neighbouring minimum frequency bins are not available. For exam-
 391 ple, the hue histogram in Fig. 5(c) is divided into two parts. Part 1 is
 392 between Min 1 to Min 2, but Part 2 is from Min 2 to Max 2 since there is
 393 no Min 3.

394 The histogram parts are sorted in descending order based on their total
 395 number of points, or total frequencies, t_i , where $1 \leq i \leq n$. Starting from
 396 the part that has the highest total points $\max(t_i)$, points are accumulated
 397 for each part from its maximum bin towards its minimum bins, adding the
 398 next largest bin at a time. The accumulation stops for a part if the number
 399 of accumulated points is at least 97% of this part and *low* and *high* thresh-
 400 old values are recorded at stop positions. In this way threshold pairs are
 401 estimated for other parts iteratively and the iteration terminates if the total
 402 number of points of the already used parts is at least 90% of $ABCD$. This
 403 means parts having very low t_i values are not considered, which helps to
 404 avoid extension of an initial position towards a vegetation area whose small
 405 region is on the roof and within the initial position, but the major region is
 406 outside the building.

407 3.4.2. Extending Initial Positions

408 The initial building positions $B_{ini} = \{b_{ini,i}\}$ are sorted in descending or-
 409 der of their *length* or *area*, since both of these sorted lists were found to
 410 offer the same performance. Then in order to obtain final building positions
 411 $B_{fin} = \{b_{fin,i}\}$, all initial positions are extended one after another, starting
 412 from the one having the longest *length* or largest *area*.

413 To extend an initial position $b_{ini,i}$ denoted by a rectangle $ABCD$, its
 414 four sides are extended separately. To extend a side, say AB , a rectangle

415 $ABNM$, with *length* $L_f = |AB|$ and *width* $W_f = 0.35m$, opposite to P_{in} , is
 416 considered. For $ABNM$ the percentages of Y , I and Q within threshold pairs
 417 $T_Y = [l_Y, h_Y]$, $T_I = [l_I, h_I]$ and $T_Q = [l_Q, h_Q]$, respectively, are computed. Let
 418 these percentages be λ , χ and μ . The percentages of black pixels in the
 419 primary and secondary building masks for $ABNM$ are also computed. Let
 420 these percentages be ς and ν . If λ , χ and μ are above 40% and ν is above 90%,
 421 AB is extended by replacing M by A and N by B . This extension procedure
 422 of AB continues iteratively and in each iteration the value of ς is checked
 423 and it should either be the same as or less than in the previous iteration. If ς
 424 becomes below 10% there is a high probability that the extension procedure
 425 will soon end. However, if ς starts increasing thereafter it is the position
 426 where AB is being extended over a neighbouring object, either a building
 427 or a tree. If this is the case, the extension procedure for AB immediately
 428 terminates. Otherwise, if any other condition fails, for example, if any of λ , χ
 429 and μ is below 40% or ν is below 90%, W_f is divided by 2 and the extension
 430 of AB continues. The procedure finally terminates if W_f is less than the
 431 image ground resolution (i.e., $< 0.1m$ in this investigation). After extension
 432 of all four sides of $ABCD$, the extended rectangle is obtained. Fig. 6 shows
 433 the final detected buildings in four tested scenes.

434 4. Performance Evaluation

435 The proposed threshold-free evaluation system makes *one-to-one* corre-
 436 spondences using nearest centre distances between detected and reference
 437 buildings. The reference buildings are obtained using manual measurement
 438 from the orthoimagery (Section 4.1). Altogether 15 indices are used in three

categories (object-based, pixel-based and geometric) to evaluate the performance. Most of these have been adopted from the literature and the rest are proposed for a more complete evaluation (Section 4.2). Section 4.3 details the experimented data sets and Section 4.4 presents results and a discussion.

4.1. Evaluation System

For evaluation, two sets of data were used, in which each building is represented either as a rectangular entity, for ‘I’ shape building, or a set of rectangular entities, for ‘L’, ‘U’ and ‘C’ shapes. The first set $B_d = \{b_{d,i}\}$, where $0 \leq i \leq m$ and m is the number of detected rectangular entities, is known as the *detected set*. It is obtained from the proposed automatic building detection technique. Each entity $b_{d,i}$ is an array of four vertices and the centre (intersection of two diagonals) of a rectangular detected entity. The second set $B_r = \{b_{r,j}\}$, where $0 \leq j \leq n$ and n is the number of reference entities, is termed the *reference set*. It is obtained from manual building measurement within the orthoimagery. Each entity $b_{r,j}$ is an array of four vertices and the centre of the rectangular reference entity.

To find the reference set B_r , manual image measurement is used. Any building-like objects above the height threshold T_h (Section 3.2) are included in B_r . As a result some garages (car-ports) whose heights are above T_h are also included, but some building parts (verandas) whose heights are below T_h are excluded. Different building parts are referred to separate rectangular entities. Consequently, there is one entity for ‘I’ shape, two entities for ‘L’ shape, three entities for ‘U’ shape, four entities for ‘C’ shape and so on.

It is natural that different rectangular entities of the same building overlap each other. In B_r , two overlapping entities must always belong to the

464 same building and represent two connected building parts (Fig. 7(a)). Such
 465 an overlap is defined as a *natural overlap* and for identification purposes a
 466 *building identification number* b_{id} is assigned to each reference entity, this
 467 being stored in $b_{r,j}$, in addition to the four vertices. Entities of the same
 468 building are assigned the same b_{id} , but those of the different buildings are
 469 assigned different b_{id} values.

470 In B_d , the situation is different. Here two overlapping entities may belong
 471 to the same building and represent two connected building parts. In such a
 472 case, this overlap is a *natural overlap* (Fig. 7(a)) and it is not counted as an
 473 error in the proposed evaluation. In all other cases, the overlap is counted
 474 as an error in the evaluation system. For example, the overlapping entities
 475 may represent the same building (multiple detection, Fig. 7(b)) or constitute
 476 combinations of true and false detections (Figs. 7(c)-(e)).

477 In an approach similar to that of Song and Haithcoat (2005), a detected
 478 entity is counted as correct if any of its part overlaps a reference entity. How-
 479 ever, unlike existing evaluation systems (Rottensteiner et al., 2005; Rutzinger
 480 et al., 2009), a *pseudo one-to-one* correspondence is established between the
 481 detected and reference sets without using any thresholds. *Pseudo one-to-one*
 482 correspondence means that each entity in one set has at most one correspon-
 483 dence in the other set. If a detected entity overlaps only one reference entity
 484 which is not overlapped by any other detected entity, then a true correspon-
 485 dence is established between them. If a detected entity overlaps more than
 486 one reference entity, then the nearest reference entity (based on the distance
 487 between centres) is considered as a true correspondence for the detected en-
 488 tity. The same rule is applied when a reference entity is overlapped by more

489 than one detected entity. As a consequence, there will be no correspondence
490 for *false positive* and *false negative* entities.

491 4.2. Evaluation Indices

492 Altogether, 15 performance evaluation indices in three categories have
493 been adopted: object-based evaluation, area- or pixel-based evaluation and
494 geometric evaluation. For pixel-based evaluation, pixels in the orthoimage
495 are used for all detected and reference entities. The geometric evaluation is
496 separated from the other two as such a evaluation estimates the positional
497 accuracy and counts neither the number of objects nor the number of pix-
498 els. In the following subsections, different indices in the three categories
499 are discussed. Note that the definitions of *true positive* (TP), *true negative*
500 (TN), *false positive* (FP) and *false negative* (FN) have been adopted from
501 Lee et al. (2003). In addition, a new term *multiple detection* (MD), which
502 indicates that for an entity presented in the reference set there are two or
503 more entities in the detected set, has also been used.

504 4.2.1. Object-based Indices

505 The following seven indices are used for object-based evaluation to evalu-
506 ate the number of buildings counted. Completeness C_m , also known as *detec-*
507 *tion rate* (Song and Haithcoat, 2005) or *producer's accuracy* (Foody, 2002),
508 *correctness* C_r , also known as *user's accuracy* (Foody, 2002) and *quality* Q_t
509 have been adopted from Rutzinger et al. (2009). The remaining four are
510 defined as:

1. *Multiple detection rate* is the percentage of multiply and correctly de-
tected entities in the detected set. As shown in Fig. 7(b), a building or

a building part can be detected more than once and all these detected entities correspond to a single entity in the reference set. The closest detected entity with respect to the reference entity is marked as a TP and all others as MDs. The *multiple detection rate* is defined as

$$M_d = \frac{|MD|}{|TP| + |FP| + |MD|}, \quad (1)$$

511 where $|\cdot|$ denotes the set cardinality. Note that $|TP| + |FP| + |MD|$
512 denotes the total number of entities in the detected set.

2. *Detection overlap rate* is the percentage of overlap in the detected set. It is defined as

$$D_o = \frac{O_d}{|TP| + |FP| + |MD|}, \quad (2)$$

513 where O_d is the number of detected entities that overlap other detected
514 entities. However, the natural overlaps are excluded.

3. *Detection cross-lap rate* is defined as the percentage of detected entities which overlap more than one reference entities and expressed as:

$$C_{rd} = \frac{C_{ld}}{|TP| + |FP| + |MD|}, \quad (3)$$

515 where C_{ld} is the number of detected entities which overlap more than
516 one reference entity and the natural overlaps are again excluded.

4. *Reference cross-lap rate* is defined as the percentage of reference entities which are overlapped by more than one detected entity and this is expressed as

$$C_{rr} = \frac{C_{lr}}{|TP| + |FN|}, \quad (4)$$

517 where C_{lr} is the number of reference entities which are overlapped by
518 multiple detected entities, with the natural overlaps being excluded.

519 A good building detection system should have high C_m and C_r values,
 520 but low M_d , D_o , C_{rd} and C_{rr} values, while $1 - C_r$ indicates the false alarm
 521 rate of the system. Q_l makes a compromise between C_m and C_r (Heipke
 522 et al., 1997).

523 4.2.2. Pixel-based Indices

524 For area- or pixel-based evaluation, pixels in the orthoimage are used for
 525 all detected and reference entities. For an FP detected entity, all the pixels
 526 within it are FP_p pixels (subscript p stands for pixels). For an FN reference
 527 entity, all the pixels within it are FN_p pixels. For a TP detected entity,
 528 there are two types of pixels: all the pixels within it that also appear in the
 529 corresponding reference TP entity are TP_p pixels and the rest are FP_p pixels.
 530 Similarly, for a TP reference entity, there are two types of pixels. All the
 531 pixels within it that also appear in the corresponding detected TP entity are
 532 TP_p pixels (counted only once) and the rest are FN_p pixels with all other
 533 pixels being TN_p pixels. Note that within the natural overlapping area the
 534 pixels are counted only once though they may be detected twice (as they
 535 are common to two detected entities on the same building). MDs are not
 536 considered in the pixel-based evaluation.

537 A total of 7 pixel-based evaluation indices are used, these being: *com-*
 538 *pleteness* C_{mp} , also known as *matched overlay* (Song and Haithcoat, 2005)
 539 and *detection rate* (Lee et al., 2003), *correctness* C_{rp} and *quality* Q_{lp} from
 540 Rutzinger et al. (2009); *area omission error* A_{oe} and *area commission error*
 541 A_{ce} from Song and Haithcoat (2005) and *branching factor* B_f and *miss factor*
 542 M_f from Lee et al. (2003). A good building detection system should have
 543 high C_{mp} and C_{rp} values, but low A_{oe} , A_{ce} , B_f and M_f values, while $1 - C_{rp}$

544 indicates the false alarm rate of the system with respect to the building area.
545 Q_{lp} makes a compromise between C_{mp} and C_{rp} (Heipke et al., 1997).

546 4.2.3. Geometric Index

547 As the geometry of the actual and detected buildings often differs signifi-
548 cantly and the generally lower spatial resolution of the LIDAR data prohibits
549 geometrically accurate building detection, the geometric evaluation system
550 is rarely found in the literature. The shape similarity indices presented in
551 Song and Haithcoat (2005) fall into this category and are application specific,
552 for example, for cadastral management. Since it is assumed that both the
553 reference and detected entities are rectangular, local changes in shapes are
554 avoided and the shape indices are not considered.

555 Song and Haithcoat (2005) utilized root-mean-square-error (RMSE) val-
556 ues in order to estimate the geometric positional accuracy. For each *one-to-*
557 *one* correspondence between detected and reference set, RMSE is measured
558 as the average distance between a pair of detected and reference entities.
559 Therefore, the RMSE is measured for TPs only, but not for FPs, FNs and
560 MDs.

561 4.3. Data Sets

562 The test data set employed here was captured over Fairfield, NSW, Aus-
563 tralia using an Optech laser scanner. Four sub-areas were used, the first
564 covering an area of $248m \times 210m$ (Fig. 6(a)), the second covering an area of
565 $155m \times 219m$ (Fig. 6(b)), the third covering an area of $228m \times 189m$ (Fig.
566 6(c)) and the fourth covering an area of $586m \times 415m$ (Fig. 6(d)). While the
567 first two areas contain only residential buildings, the last two areas contain

568 both residential and industrial buildings. The first three areas were used
569 for objective evaluation using 15 indices, while the fourth is a bigger area
570 containing around 400 buildings and was used for visualization only. Last-
571 pulse LIDAR data with a point spacing of $0.5m$ was used. A DEM (with $1m$
572 spacing) and four RGB colour orthophotos with a resolution of $0.15m$ were
573 available for these areas. The fact that the orthoimage did not contain an
574 infrared band was circumvented by computing a pseudo-NDVI image using
575 the assumption that the three image bands are in the order of IR-Red-Green
576 in order to be used in the standard NDVI formula (Kidwell, 1997).

577 The orthoimagery had been created using a bare-earth DEM, so that the
578 roofs and the tree-tops were displaced with respect to the LIDAR data. Thus,
579 data alignment was not perfect. Apart from this registration problem, there
580 were also problems with shadows in the orthophotos, so the pseudo-NDVI
581 image did not provide as much information as expected.

582 Reference data sets were created by monoscopic image measurement using
583 the Barista software (Barista, 2009). All rectangular structures, recognizable
584 as buildings and above the height threshold T_h (Section 4.1) were digitized.
585 The reference data included garden sheds, garages, etc., that were sometimes
586 as small as $10m^2$ in area. Altogether, 70, 62 and 60 buildings from the first
587 three scenes formed the reference sets.

588 4.4. Results and Discussion

589 The algorithm was implemented and tested using Matlab 7.8.0 (R2009a)
590 on a Windows XP machine with 3.00GHz of Intel(R) Core(TM)2 Duo CPU
591 and 3.23GB of RAM. The average running time for first three scenes was
592 about 13.5 minutes. The majority of time was taken up with accessing, load-

ing and storing the high volume of input and intermediate data. Memory-related limitations in Matlab precluded the possibility of conducting an experiment covering all 2400 buildings within the Fairfield data set. Such experimental validation will be possible once the algorithms are fully implemented within the Barista software.

The experimentation was carried out in two phases. Firstly, a sensitivity analysis of five important parameters (tile size, black pixel threshold, NDVI threshold, area reduction (reducing *length* and *width* of initial buildings) and colour similarity) was carried out to test how the detection algorithm performed when parameter values were changed. The standard parameter values were chosen for the test data sets. Secondly, the detection performance was evaluated using 15 indices in three categories when all the parameters were set at their chosen standard values.

4.4.1. Sensitivity Analysis

For sensitivity analysis five different values for each of the five parameters were used and object and pixel-based *qualities* were estimated. The reason for choosing *quality* as a measurement for sensitivity analysis is that it provides a balance between completeness and correctness (Heipke et al., 1997). The following values were used for the six parameters:

- *Tile size*: 400×400 , 450×450 , 500×500 , 550×550 and 600×600 pixels;
- *Black pixel threshold*: 0.6, 0.7, 0.8, 0.9 and 1.0;
- *NDVI threshold*: 32, 40, 48, 56 and 64;

- 616 • *Area reduction*: 0.1, 0.15, 0.2, 0.25 and 3.0 and
- 617 • *Colour similarity*: 0.2, 0.3, 0.4, 0.5 and 0.6.

618 Fig. 8, in which the numbers 1 to 5 along the x -axis indicate the five
 619 values for each parameter, graphically illustrates the results. When one of
 620 the parameters was changed, others were set at their standard values. The
 621 pixel-based quality was given more weight than the object-based quality in
 622 the choice of the standard value for each parameter. Overall, one parameter
 623 - area reduction - was found to be moderately sensitive, while the other four
 624 were found less sensitive.

625 While the highest object-based quality was achieved at 20% area reduc-
 626 tion and the highest pixel-based quality was achieved at 10% area reduction,
 627 at 15% area reduction both of these qualities were slightly lower than their
 628 highest values. Both the object- and pixel-based qualities were highest when
 629 the color similarity was 40%. For the tile size and NDVI threshold, the chosen
 630 values were 450×450 pixels and 48, respectively, when pixel-based qualities
 631 were highest and object-based qualities were slightly below the highest. An
 632 opposite scenario was observed when the black pixel threshold was 90%.

633 While a smaller tile size makes the mask generation procedure a bit ex-
 634 pensive, a larger tile size may not clearly distinguish some buildings in a
 635 sloping tile because the estimated height threshold may not perfectly sepa-
 636 rate ground and above ground objects throughout the tile. A small NDVI
 637 threshold may remove some buildings as vegetation if building roofs have
 638 colours that are similar to trees. In contrast, a large NDVI threshold may
 639 detect some trees as buildings.

Table 1: Object-based evaluation results in percentages (C_m = completeness, C_r = correctness, Q_l = quality, M_d = multiple detection rate, D_o = Detection overlap rate, C_{rd} = detection cross-lap rate and C_{rr} = reference cross-lap rate).

Scenes	C_m	C_r	Q_l	M_d	D_o	C_{rd}	C_{rr}
Scene 1	97.14	97.14	95.31	2.60	6.85	2.74	7.14
Scene 2	95.94	96.55	92.08	4.62	5.00	1.67	4.84
Scene 3	98.33	99.25	90.94	11.60	28.57	14.29	33.33
Average	97.14	97.9	92.78	6.27	13.47	6.23	15.11

640 A smaller than 90% black pixel threshold may result in a nearby tree being
 641 included as a building part, for example. The same may happen if a more
 642 than 40% colour similarity is used when the building roof has a similar colour
 643 to the tree. While a smaller than 15% area reduction (*length* and *width* of
 644 initial buildings reduced by 15%) may not fully correct the registration error,
 645 a larger area reduction may stop the extension of the initial position if the roof
 646 has slightly different colours. This is why the pixel-based quality dropped
 647 more rapidly than the object-based quality, which indicates that though the
 648 buildings are correctly detected they are not correctly delineated. This is
 649 also evident from the evaluation results discussed below.

650 4.4.2. Evaluation using Standard Parameter Values

651 Table 1 shows the object-based evaluation results and Table 2 shows the
 652 pixel-based evaluation results. The geometric accuracy (RMSE) for the three
 653 scenes was 1.98m, 1.91m and 1.86m with an average accuracy of 13 pixels
 654 (1.92m).

655 In object-based evaluation, more than 97% *completeness* and *correct-*

Table 2: Pixel-based evaluation results in percentages (C_{mp} = completeness, C_{rp} = correctness, Q_{lp} = quality, A_{oe} = area omission error, A_{ce} = area commission error, B_f = branching factor and M_f = miss factor).

Scenes	C_{mp}	C_{rp}	Q_{lp}	A_{oe}	A_{ce}	B_f	M_f
Scene 1	77.32	89.29	70.07	22.68	10.35	12.00	29.33
Scene 2	77.97	87.05	67.40	22.03	12.67	14.87	28.26
Scene 3	79.51	90.35	72.11	20.49	7.54	10.68	25.77
Average	78.27	88.90	69.86	21.74	10.19	12.52	27.79

ness resulted in an average 92% *quality* with at least 6% of buildings being detected multiple times. The *reference cross-lap rate* was higher than the *detection cross-lap rate*, since some nearby trees were detected along with the actual buildings. In pixel-based evaluation, while 78% of building areas were completely detected, resulting in a 21% *omission error*, 89% of detected areas were correct, offering a 10% *commission error*. Since the *miss factor* and *omission error* were larger than the *branching factor* and *commission error*, respectively, the *false positive rate* of the proposed technique is lower than its *false negative rate*.

Overall, in both object- and pixel-based evaluations, the proposed detection technique performed better on Scene 1 than on Scene 2 in terms of all indices except *cross-lap* and *detection overlap* rates. There were two reasons for this: a) some true buildings were detected twice in Scene 1, and b) in Scene 1 though all true buildings were detected with some of them being missed partially, some false buildings (actually trees) were also detected. Scene 3 performed better than Scenes 1 and 2 in pixel-based evaluation whereas Scene

672 3 gave higher *cross-lap* and *detection overlap rates* in object-based evaluation
673 due to multiple detection of complex industrial buildings. In the geometric
674 evaluation, in terms of RMSE, there was at least $0.05m$ better positional
675 accuracy for Scene 3 than for Scenes 1 and 2.

676 It was found that the use of NDVI (actually pseudo NDVI in this case) did
677 not perform well in distinguishing between trees and building roofs, especially
678 when both were of similar colour. While a low NDVI threshold removed some
679 true buildings, a high NDVI threshold detected some trees as buildings. The
680 difference in first- and last-pulse LIDAR data was also investigated to remove
681 trees, as was done in Rottensteiner et al. (2005), but was found less useful.
682 The outcome regarding NDVI and the difference in first and last pulse data
683 supports the finding of an earlier study by Rottensteiner et al. (2007).

684 Since different published detection techniques follow different evaluation
685 systems on different data sets, they are difficult to compare. As with the
686 proposed detection technique, Sohn and Dowman (2007) and Cheng et al.
687 (2008) also used line segments and building geometry adjustment using dom-
688 inant line angles. Unlike Cheng et al. (2008) and the proposed technique,
689 Sohn and Dowman (2007) used specific building models to fit the LIDAR
690 points. While the proposed technique introduces a threshold-free evaluation
691 system, both of these existing techniques employ threshold-based evaluation
692 systems.

693 In terms of object-based *correctness* the proposed technique performed
694 much better than that of Cheng et al. (2008). The method of Sohn and Dow-
695 man (2007) offered slightly higher pixel-based performance than the proposed
696 technique because of the adopted evaluation system (Rottensteiner et al.,

2005), which excluded FP and FN buildings from evaluation and established many-to-many relationships between the detected and reference sets. Establishing one-to-one correspondences by the proposed detector increases the number of FP and FN buildings and the proposed evaluation system considers all of them.

The same Fairfield data set was previously employed by Rottensteiner et al. (2005), Rottensteiner et al. (2007) and Rutzinger et al. (2009) to investigate automated building extraction. However, in those investigations, two different threshold-based evaluation systems were employed and the Dempster-Shafer (DS) detector was evaluated using *completeness*, *correctness* and *quality*. Rutzinger et al. (2009) has presented results of pixel-based evaluation of the DS detector showing that it can offer higher *completeness* (92.1%) and *quality* (81.8%) than the proposed detector. However, in object-based evaluation the DS detector offered much lower *completeness* (44.2%) and *quality* (43.1%) than the proposed detector. The superior performance of the DS detector in pixel-based evaluation was largely due to the adopted evaluation systems, Rottensteiner et al. (2005) and Rutzinger et al. (2009) which excluded FP and FN buildings from evaluation and established many-to-many relationships between the detected and reference sets. Moreover, unlike the proposed detector the DS detector was excessively sensitive to small buildings (performance deteriorated with the decrease of building size) and buildings smaller than $30m^2$ could not be detected (Rottensteiner et al., 2007).

720 5. Conclusion

721 This paper has proposed an automatic building detection technique using
722 LIDAR data and multispectral imagery. The initial building positions are
723 obtained from the primary building mask derived from LIDAR data. The
724 final building positions are obtained by extending their initial positions based
725 on colour information, and the two masks ensure the accurate delineation
726 of the buildings. In particular, the primary building mask helps separate
727 building detections when they are very close to each other and the secondary
728 building mask helps to confine the extension of initial positions outside a
729 building when the roof and ground have similar colour information.

730 Experimental testing has shown that the proposed technique can detect
731 urban residential and industrial buildings of different shapes with a very high
732 success rate. However, the technique can display shortcomings in areas of
733 high-terrain slope and those with dense high-rise buildings of rapidly varying
734 height within a given tile size, since in such areas the average DEM height
735 may not necessarily correspond to the actual ground height. Extension of the
736 algorithm’s functionality to better accommodate such situations is currently
737 under investigation.

738 Another important observation from the presented results is that object-
739 based *completeness* (*detection rate* 97%) is high when compared to pixel-
740 based *completeness* (*matching overlay* 78%). However, the geometric po-
741 sitional accuracy remains relatively poor (13 pixels) for mapping purposes;
742 although not for applications where building detection is the primary goal.
743 This observation indicates that some of the truly detected buildings are not
744 completely delineated due to small local variations along the roof boundary,

745 occlusion by nearby trees or different roof colours in and out of the initial
746 building position. Consequently, the proposed detection technique can be
747 applied in city planning, homeland security, disaster (flood or bushfire) man-
748 agement and building change detection with high reliability, but it is not as
749 yet applicable to cadastral mapping and accurate roof plane extraction, both
750 of which require higher pixel-based and geometric accuracy.

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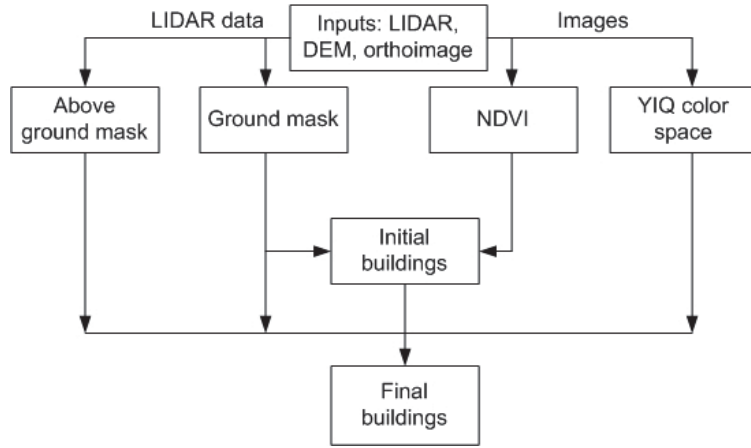


Figure 1: Flow diagram of the proposed building detection technique.

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838 metry, Remote Sensing and Spatial Information Sciences 37 (part B1),
839 197–202.

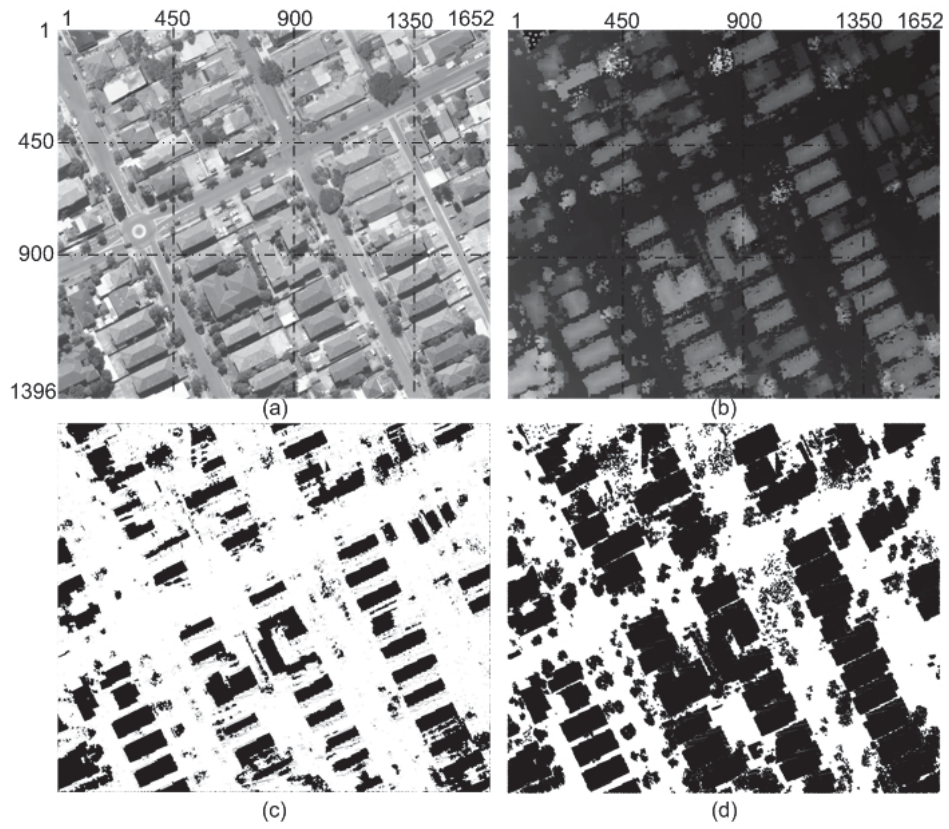


Figure 2: (a) A test scene, (b) LIDAR data (shown in gray-scale), (c) primary building mask and (d) secondary building mask.



Figure 3: (a) Application of NDVI and (b) line-adjustment to remove tree-edges. Lines with small circles at centres are removed. (c) Initial building positions.

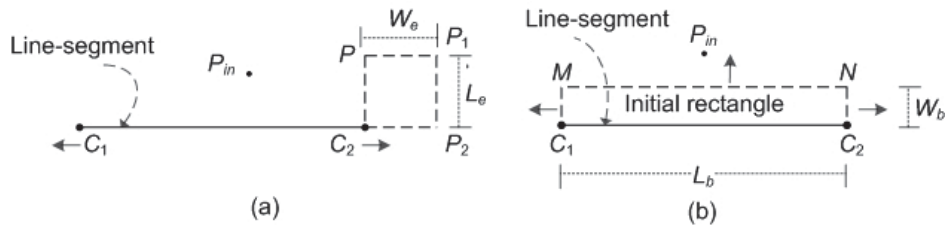


Figure 4: (a) Extending a line segment (b) Forming an initial building by extending three sides of a rectangle on a line segment. Arrows indicate extension directions.

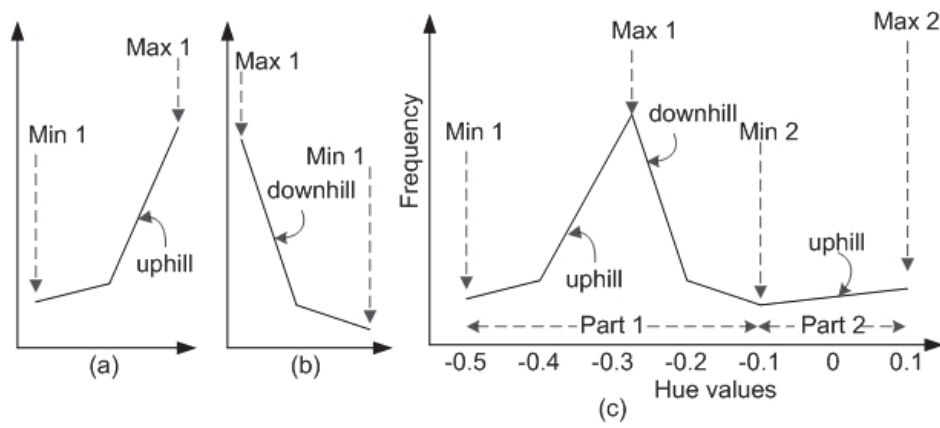


Figure 5: Basic histograms: (a) uphill and (b) downhill. (c) a combination of basics uphill-downhill-uphill.



Figure 6: Detected buildings on the orthoimages.

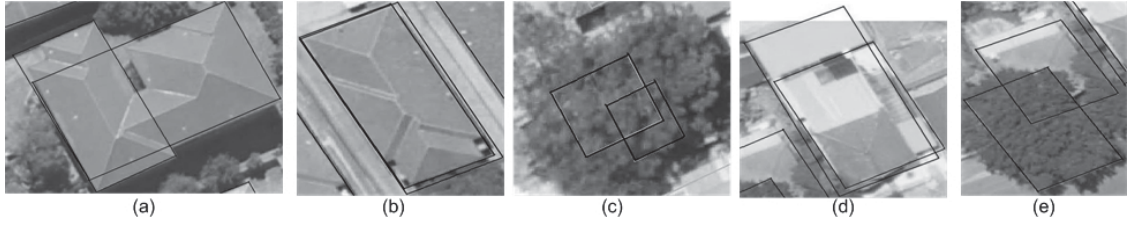


Figure 7: Different types of detection overlaps: (a) natural, (b) multiple detection, (c) false-false, (d) true-true and (e) true-false.

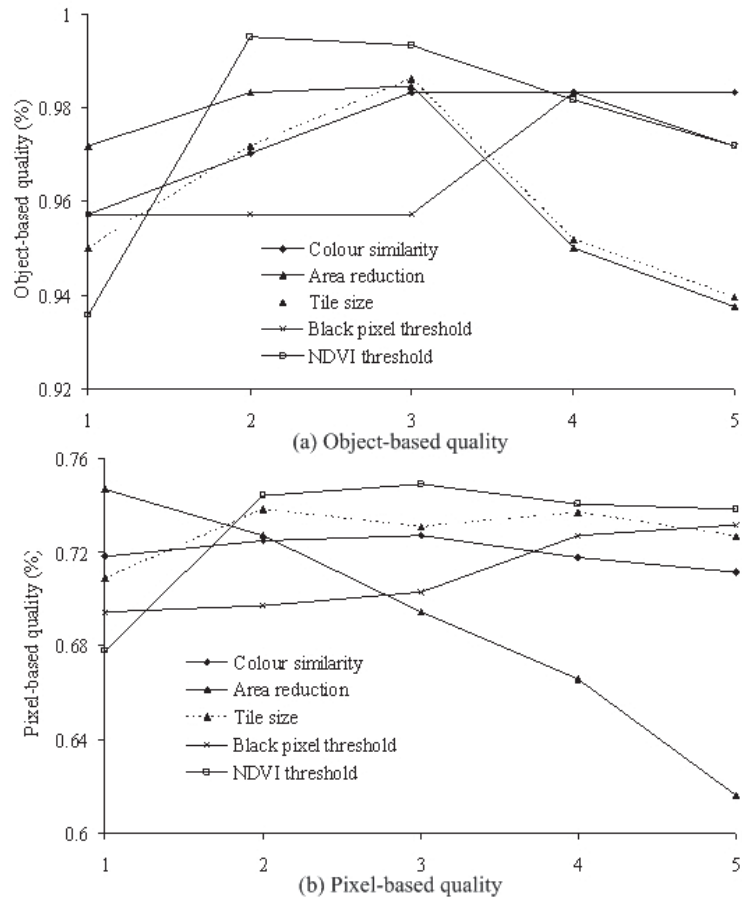


Figure 8: Sensitivity of different parameters.